Answer Key — Reaction Intermediates

Short answers only; full reasoning is in the book. Answers written for this book have not been independently checked; ⚑ marks convention-dependent items.

Answer Key All 552 short answers in book order

ID	Answer
E2-1	3 > 2 > 1
E2-2	1 > 3 > 2
E2-3	1 > 2 > 3
E2-5	1 > 2 > 3
E2-6	3 > 2 > 1
E2-7	1 > 2 > 3
E2-8	3 > 2 > 1
E2-9	1 > 2 > 3
E2-10	1 > 2 > 3
E2-11	1 > 2 > 3
E2-12	1 > 3 > 2
E2-13	2 > 3 > 1
E2-14	3 > 2 > 1
E2-15	1 > 2 > 3
E2-16	3 > 2 > 1
E2-17	1 > 2 > 3
E2-20	3 > 2 > 1
E2-21	1 > 2 > 3
E2-23	2 > 1 > 3
E2-24	3 > 2 > 1
E2-25	3 > 2 > 1
E2-26	4 > 2 > 3 > 1
E2-27	4 > 2 > 3 > 1
E2-28	3 > 2 > 1 > 4
E2-29	3 > 1 > 2
E2-30	3 > 1 > 2
E2-31	1 > 3 > 2
E2-32	1 > 3 > 2
E2-35	1 > 2 > 3
E2-91	(c)

E2-93	(c)
E3-8	(D) neopentyl
E3-9(i)	c > b > a
E3-9(ii)	c > b > a
E3-9(iii)	b > c > a (key)
E3-9(vi)	key: b > c > a; your Allen solution page: a > b > c
E3-9(iv)	d > c > b > a (textbook key)
E3-9(v)	c > b > a
E3-9(vii)	c > b > a
E3-9(viii)	c > a > d > b
E3-9(ix)	b > c > a
E3-9(x)	c > a > b
E3-9(xi)	c > a > b
E3-9(xii)	a > b > c > d
E3-9(xiii)	b > a > c > d
E3-9(xv)	a > c > b
E3-9(xvi)	b > c > a
E3-10	(C)
E3-11a/b	(a) benzyl; (b) ethyl
E3-13	(A)
E3-14	d < b < c < a → (A)
E3-15	2 > 1 > 3 > 4 → (D)
E3-16	II < III < I → (B)
C-7	3 (p-OCH_3, p-OH, p-NH_2)
C-11a	3 (B, D and E)
C-12a	3 (B, C and J)
C-11b	x = 4, y = 2, value = 21/7 = 3
C-12b	3 (iii, iv, vi)
C-15	3 (i, ii, iii)
C-23	7 (i, iii, vii, ix, x, xi, xii)
D-27	4 (I, II, IV, V)
D-38	5 (i, ii, iii, iv, vi correct; v is wrong)
D-19 / D-21b	4 (d, f, i, k)

D-21a	4 (1, 6, 7, 8)
A-48	(a) i > ii > iii
A-49	(b) ii > iii > i
A-50	(c) iii > ii > i
A-51	(b) i > ii > iv > iii
SQ-1	d > c > b > a
SQ-2	a > b > c > d > e
SQ-3	d > c > b > a
SQ-4	a > b > c
SQ-5	a > b > c > d
SQ-6	c > b > a
SQ-7	pentadienyl
SQ-8	Ph_2CH^+
SQ-9	$(p\text{-}MeOC_6H_4)_3C^+ > Ph_3C^+ > (p\text{-}O_2NC_6H_4)_3C^+$
SQ-10	a > b > c > d
SQ-11	a > b > c
SQ-12	the β-silyl cation
SQ-13	a > b > c
SQ-14	(a) 4; (b) 7; (c) 10
SQ-15	2-bromo-2-methylbutane; 2° cation → 1,2-H shift → 3° cation
SQ-16	2-chloro-2,3-dimethylbutane
SQ-17	2-bromo-2-methylbutane
SQ-18	1-bromo-1-methylcyclohexane; primary → tertiary by 1,2-H shift
SQ-19	2-methylbut-2-ene
SQ-20	$CH_3OCH_2^+$
SQ-21	$t\text{-}Bu^+$
SQ-22	$PhCH(CH_3)^+$
SQ-23	o-methoxybenzyl
SQ-24	p-methylbenzyl
SQ-25	$m\text{-}CH_3 > H > m\text{-}OCH_3 > m\text{-}Cl > m\text{-}NO_2$
SQ-26	p-F > H > p-Cl > p-Br
SQ-27	dianisyl > Ph_2CH^+ > dinitrophenyl
SQ-28	$(c\text{-}C_3H_5)_3C^+ > (c\text{-}C_3H_5)_2CH^+ > c\text{-}C_3H_5CH_2^+$
SQ-29	para ≈ ortho > meta
SQ-30	para ≥ ortho ⋙ meta
SQ-31	meta > ortho ≈ para
SQ-32	a > b > c
SQ-33	HIA: CH_3^+ (312) > $C_2H_5^+$ (270) > $i\text{-}Pr^+$ (249) > $t\text{-}Bu^+$ (231 kcal mol^{-1}); lowest is $t\text{-}Bu^+$
SQ-34	⚑ it depends on the phase and the source
SQ-35	σ-participation by the C1–C6 bond from the back side of the leaving group (only possible for exo)
SQ-36	(i) false (6); (ii) true; (iii) false; (iv) true
SQ-37	(a) 9; (b) 6; (c) 3; (d) 8

SQ-38	a > b > c
SQ-39	$(CH_3)_2C{=}O^+CH_3$
SQ-40	⚑ benzyl > cyclopropylcarbinyl > allyl (most JEE keys); gas-phase data put allyl lowest, benzyl and cyclopropylcarbinyl close
SQ-41	2-bromo-2-methylbutane
SQ-42	(b) → methyl shift to 2,3-dimethylbut-2-yl; (c) → hydride shift to 2-methylbut-2-yl (3°)
SQ-43	t-Bu^+ (from isobutene) > i-Pr^+ (propene) > ethyl (ethene)
SQ-44	the C=O is an electron-withdrawing group (–I and –M) next to an electron-deficient carbon
SQ-45	t-BuCl > i-PrCl > EtCl ≳ MeCl (≈ 10^8 : 45 : 1 : 1)
SQ-46	the isopropyl cation (from propene + H^+) attacks benzene (Friedel–Crafts alkylation)
SQ-47	hydride shift from the CH_2Cl-bound complex gives the secondary cation
SQ-48	⚑ i-PrCl < allyl Cl < t-BuCl ≈ benzyl chloride; commonly accepted order i-PrCl < allyl < benzyl < t-BuCl
SQ-49	$CH_3CH_2^+ > CH_2{=}CH^+ > HC \equiv C^+$
SQ-50	π-participation by the C2=C3 bond gives a stabilised bridged (homoallylic) cation
MA-1	phenyl migrates: 3,3-diphenylbutan-2-one
MA-2	diphenylacetaldehyde
MA-3	isobutyraldehyde (2-methylpropanal)
MA-4	phenylacetaldehyde
MA-5	2-methylcyclohexanone
MA-6	deoxybenzoin ($PhCOCH_2Ph$, 1,2-diphenylethanone)
MA-7	cycloheptanone
MA-8	(a) t-butyl; (b) phenyl; (c) cyclohexyl; (d) benzyl
MA-9	cyclopentyl bromide (ring expansion)
MA-10	methyl shift gives a tertiary cation with no strain change
MA-11	the product is triphenylacetophenone; phenyl migrates because OH cannot, and Ph is the only migrating candidate
MA-12	phenyl (to give 3-methyl-4-phenylphenol)
MA-13	2,3-dimethylbut-2-ene; methyl migrates
MA-14	isobutyraldehyde (Meinwald rearrangement)
MA-15	phenyl (it is anti to OH): acetanilide
MA-16	H > p-$MeOC_6H_4$ > Ph > t-Bu > CH_3 (⚑ H vs aryl)
MA-17	p-anisyl > phenyl > p-chlorophenyl > p-nitrophenyl
MA-18	the more substituted (secondary C-2) carbon: 7-methyloxepan-2-one
MA-19	retention
MA-20	the shift is concerted with departure of the leaving group
MA-21	ring expansion
MA-22	aryl π electrons give a stabilised bridged (phenonium or π-complex) transition state
MA-23	the OH on the diaryl carbon leaves; 3,3-diphenylbutan-2-one
MA-24	isopropyl
MA-25	false
CA-1	a > b > c > d
CA-2	c > b > a

CA-3	a > b > d > c (JEE-type key; DMSO pK_a: 17.2, 26.5, 31.3, ≈ 30)
CA-4	(b) and (d)
CA-5	aromatic stabilisation of the anion
CA-6	a > b > c > d (≈ 16, 20, 22.6, 36)
CA-7	(A)
CA-8	attack at the β-carbon gives the enolate
CA-9	$^{-}CCl_3$ then $:CCl_2$
CA-10	methane via CH_3^-
CA-11	planar enolate (carbanion)
CA-12	(i) 2,6-dimethylcyclohexanone; (ii) 2,2-dimethylcyclohexanone
CA-13	2
CA-14	(b) and (e)
CA-15	A–R; B–Q; C–P; D–S
CA-16	3 (ethyl acetoacetate, diethyl malonate, nitromethane)
CA-17	$K = 10^{-3.4} \approx 4 \times 10^{-4}$
CA-18	(a) 2,6-dimethyl; (b) 2,2-dimethyl; (c) 2,2-dimethyl
CA-19	Meldrum's acid, marginally; both far stronger than diethyl malonate
CA-20	ethyl acetoacetate; pivalate has no α-H; isobutyrate gives a product without an acidic H
CA-21	(a) benzalacetone; (b) chalcone; (c) 2-benzylidenecyclohexanone
CA-22	4
CA-23	(a) $CHI_3 + CH_3COO^-$; (b) $CHCl_3 + C_6H_5COO^-$; (c) $CHI_3 + HCOO^-$; yellow CHI_3 from (a) and (c)
CA-24	(a) 1; (b) 1
CA-25	c < a < d ≈ e < b (aqueous pKa ≈ 25, 19.3, 10.2, 11.2, 8.9)
CA-26	central position; 3-methylpentane-2,4-dione
CA-27	(a) mainly (Z)-1-phenylpropene; (b) (E)-ethyl cinnamate
CA-28	$Ph_3P{=}CH_2$ (Wittig)
CA-29	planar enolate; rate = k[ketone][HO^-], independent of any halogen
CA-30	(a) 1-methylcyclohexanol (C=O attack); (b) 2-methylcyclohexanone (Stork enamine alkylation at α-C); (c) 2-methylcyclohexanone; (d) mixture up to 2,2,6,6-tetramethylcyclohexanone
CA-31	see reasoning
CA-32	d > c > b > a > e (acid pK_a ≈ 16, 20, 31, 41, 50)
CA-33	(B) and (D)
CA-34	7,7-dichlorobicyclo[4.1.0]heptane; the catalyst carries HO^- (or CCl_3^-) into the organic phase
CA-35	n-BuLi/TMEDA only (pK_a of benzene ≈ 43)
CA-36	no, racemic
CA-37	(c) does not give the clean alcohol
CA-38	(i) aldol then E1cB; (ii) crotonaldehyde; (iii) acrolein or (with excess HCHO) pentaerythritol precursors; (iv) cinnamic acid (Perkin)
CA-39	incorrect
CA-40	no β-H in $CHCl_3$; the anion CCl_3^- is stable and eliminates from the same carbon
CA-41	the syn aldol

CA-42	cyclopentadiene; at the CH_2 (C-7)
CA-43	A: less substituted (kinetic, LDA −78 °C); B: more substituted (thermodynamic, NaOEt/EtOH)
CA-44	OR lone-pair donation into C=O
CA-45	6 (CH_3OH, CH_3COOH, CH_3CONH_2, HC ≡ CH, $C_6H_5NH_2$, H_2O)
CA-46	each fused benzene ring competes for the π electrons
CA-47	4 and 2
CA-48	benzoylacetaldehyde (3-oxo-3-phenylpropanal), largely as its enol
CA-49	ethyl 3-oxo-5-phenylpentanoate (γ-alkylation)
CA-50	c < b < a < d
CA-52	$\Delta^{1(9)}$-octalone (4,4a,5,6,7,8-hexahydronaphthalen-2(3H)-one); Michael, intramolecular aldol, E1cB dehydration
CA-53	diethyl cyclopropane-1,1-dicarboxylate; cyclopropanecarboxylic acid
CA-54	2-ethoxycarbonyl-4-methylcyclopentanone (major) over the 3-methyl isomer
CA-55	2-nitro-1-phenylethanol; β-nitrostyrene
CA-56	(a) 3-methylpentan-3-ol; (b) 2-methylbutan-2-ol; different
CA-57	5
CA-58	1-phenylbut-3-en-1-ol via [2,3]-Wittig rearrangement
CA-60	acetylacetone (9) < nitromethane (10) < diethyl malonate (13) < acetone (19) < ethyl acetate (≈ 25)
CA-61	(a) 2-benzyl-6-methylcyclohexanone (kinetic enolate); (b) 2-benzyl-2-methylcyclohexanone (thermodynamic enolate)
CA-62	(A)
CA-63	P–3, Q–2, R–4, S–1
CA-64	(E)-ethyl cinnamate and (Z)-1-phenylpropene; $Ph_3P{=}O$
CA-65	9
CA-66	(b) syn; (c) one racemic diastereomer (2 enantiomers); E-enolate gives the anti aldol
CA-67	A: ethyl 2-oxocyclopentanecarboxylate; B: its 1-methyl derivative; C: 2-methylcyclopentanone
CA-68	p = 9; log K_2 = +2
CA-69	A: $PhCOCH_2CH_2NMe_2$; B: phenyl vinyl ketone; C: diethyl 2-(3-oxo-3-phenylpropyl)malonate; E1cB
CA-70	less substituted enamine; methyl 3-(3-methyl-2-oxocyclohexyl)propanoate (alkylation at C-6)
CA-71	A, B and C
CA-72	A: ethyl 1-oxaspiro[2.5]octane-2-carboxylate (glycidic ester); B: cyclohexanecarbaldehyde
CA-73	cyclopropanone; label found about 1:1 at ring C-1 (bearing CO_2Me) and ring C-2; semibenzilic would give label only at C-1
CA-74	ethyl 2-oxo-2H-chromene-3-carboxylate (ethyl coumarin-3-carboxylate)
CA-75	[2,3]-Wittig: 2-methyl-1-phenylbut-3-en-1-ol; minor from [1,2]-Wittig; 4 stereoisomers
CA-76	P = $Ph_2CH{-}OSiMe_3$ ([1,2]-Brook); Q = $PhCH(OH)SiMe_3$ ([1,2]-retro-Brook)
CA-77	1,1,2-triphenylethane (via $Ph_2C(Li)CH_2Ph$); neophyllithium does not
CA-78	2 new C–C bonds; 1 stereocentre (quaternary C-CH_3); product racemic with achiral base
CA-79	X = hexane-2,5-dione; Y = 3-methylcyclopent-2-en-1-one; heptane-2,6-dione gives 3-methylcyclohex-2-en-1-one

CA-80	2-methylpent-4-enoic acid (via silyl ketene acetal, Ireland–Claisen [3,3]); crotyl gives 2,3-dimethylpent-4-enoic acid (two diastereomers selectively)
CA-81	vinyllithium; 3-methylcyclohexene; 6-methylcyclohex-1-ene-1-carbaldehyde
FR-1	d > e > c > b > a > f
FR-2	a > b > c > d > e > f
FR-3	Cl: 56 %; Br: 96.5 %
FR-4	(B) isobutyl chloride (≈ 64 %)
FR-5	4; 6
FR-6	(c), (d), (e)
FR-7	thermochemistry of the two propagation steps
FR-8	3-bromobut-1-ene and 1-bromobut-2-ene
FR-9	(i) benzyl chloride (then benzal chloride, benzotrichloride); (ii) o- and p-chlorotoluene
FR-10	(i) n-butane; (ii) ethane, propane, n-butane (≈ 1 : 2 : 1)
FR-11	(b) 1,2-aryl; (d) is a ring-opening (β-scission), not a shift
FR-12	delocalisation + steric protection
FR-13	III > II > I (Nucleus Q40)
FR-14	A–Q; B–P; C–R; D–S
FR-15	primary kinetic isotope effect
FR-16	racemic bromide
FR-17	27.8 % : 23.1 % : 35.2 % : 13.9 %
FR-18	4 constitutional isomers, 6 including stereoisomers; 2 of the constitutional isomers are chiral
FR-19	late versus early transition state
FR-20	only HBr has two exothermic propagation steps
FR-21	(a) 3-bromocyclohexene; (b) benzyl bromide; (c) 3-bromo-1-pentene and 1-bromo-2-pentene (mixture); (d) allyl bromide; (e) t-butyl bromide; (f) 1-bromo-1-methylcyclohexane
FR-22	$PhCH_2CH_2Br$: one carbon less than the acid
FR-23	b < e < a < f < d ≈ c (⚑ d vs c)
FR-24	2,3-dimethylbutane; $(CH_3)_2CHCH(CH_3)_2$: 12 primary H : 2 tertiary H → 12 × 1 : 2 × 5 = 12 : 10
FR-25	racemic
FR-26	Br• adds reversibly, the radical rotates, and Br• is eliminated
FR-27	(a) 1-bromopropane; (b) $CF_3CH_2CH_2Br$; (c) 2-bromopropane for propene (ionic)
FR-28	$R_p = k_p[M](f\,k_d[I]/k_t)^{1/2}$
FR-29	(a) 1000; (b) 500
FR-30	benzyl bromide (side chain, radical); o- and p-bromotoluene (ring, ionic); benzyl bromide
FR-31	(a) ethane; (b) ethane + propane + butane (1 : 2 : 1 by statistics); (c) ethene
FR-32	5 constitutional isomers; the tertiary bromide
FR-33	(a) 1-bromobutane; (b) 2-bromobutane; (c) 1-bromo-2-butene (mainly) and 3-bromo-1-butene
FR-34	Ph, vinyl, acyl, Cl/Br can; H and alkyl cannot
FR-35	cumene hydroperoxide, then phenol + acetone (Hock process)
FR-36	radical concentration is ≈ 10^{-8} M
FR-37	(b) hydroquinone

FR-38	10^4; the reaction is inhibited (induction period)
FR-39	(c) TEMPO is stable (isolable); (b) trityl persists in solution but dimerises on concentration
FR-40	(i) chlorobenzene (Gattermann); (ii) biphenyl (Gomberg–Bachmann); (iii) butane (Kolbe)
FR-41	6
FR-42	5-exo (cyclopentylmethyl) major; 6-endo (cyclohexyl) minor
FR-43	radical reduction proceeds; S_N1 does not
FR-44	benzylic H is 10^3 times more reactive than the terminal CH_3 H
FR-45	$\Delta H(Cl) = 105 - 103 = +2$ (+1); $\Delta H(Br) = 105 - 87.5 = +17.5$
FR-46	5 (ethane, neopentane, cyclohexane, toluene, 2,2,3,3-tetramethylbutane; adamantane gives two isomers, tertiary and secondary; benzene does not react by radical substitution)
FR-47	$\approx$ 90.5 %
FR-48	step 1 +2; step 2 −27; overall −25 kcal mol^{-1}
FR-49	(2-bromoethyl)benzene via the benzylic radical
FR-50	R–H (deoxygenation); formation of a strong C=O (and Sn–S) bond
FR-51	(a) methylcyclopentane; (b) hex-1-ene
FR-52	+9 (Br•); −6.5 (Cl•)
FR-53	$\nu = 10^4$; (a) 2×10^4; (b) 10^4
FR-54	3-bromo-1,1,1-trichlorononane
FR-55	N-methylpyrrolidine (Hofmann–Löffler–Freytag)
FR-56	it donates its O–H hydrogen to ROO•; the aryloxyl radical is stabilised and hindered
FR-57	4
FR-58	R–I > R–Br > R–Cl
FR-59	(a) 4; (b) $\approx$ 35 %; (c) $\approx$ 4.0 kJ mol^{-1}
FR-60	Cl: +8, −108, overall −100; Br: +73, −100, overall −27 kJ mol^{-1}; ratio $\approx 4 \times 10^{-8}$
FR-61	3 (1,1-, 1,3- and 1,4-dichloro-2-methylbutane)
FR-62	4
FR-63	A, C, D, F
FR-64	1-bromo-2-methylcyclohexane; 4 stereoisomers; mainly trans (anti addition)
FR-65	only HBr; $\Delta H(\text{overall}) = -64$ kJ mol^{-1}
FR-66	1,1,1,3-tetrachlorononane; $\approx$ 94 % 1:1 and $\approx$ 5.5 % 2:1
FR-67	$[P\bullet] = 3.2 \times 10^{-8}$ M; $R_p = 4.3 \times 10^{-5}$ M s^{-1}; $\nu \approx 358$; $X_n \approx 717$; $M_n \approx 7.5 \times 10^4$; t $\approx$ 2.7 h
FR-68	(a) $\approx$ 130; (b) $\approx 5.3 \times 10^{-3}$ M; (c) R_p unchanged; (d) $\approx$ 833 s (14 min)
FR-69	(a) $F_1 \approx 0.51$; (b) $f_1 \approx 0.53$; (c) 1.52; $r_1 r_2 = 0.24$, tendency towards alternation
FR-70	2 ((b) and (c)); (d) gives mainly cis-1,2-dimethylcyclopentane
FR-71	tert-butylbenzene (unrearranged) and isobutylbenzene (rearranged); (b) $\approx$ 0.32 %; (c) $\approx 3.2 \times 10^{-4}$ M
FR-72	hept-1-ene and 1,2-dimethylcyclopentane (cis major); 91 % cyclised at 0.010 M, 9.1 % at 1.0 M; Bu_3Sn–S–C(=O)SMe by-product
FR-73	methyl 4,4-dimethylpentanoate; $\approx$ 96 %; polarity matching (SOMO–LUMO / SOMO–HOMO)
FR-74	(b) −4 kcal mol^{-1}; (c) $\approx$ 666 g phenol, $\approx$ 411 g acetone; (d) $\approx 1.0 \times 10^5$ s ($\approx$ 28 h)
FR-75	P–2, Q–3, R–1, S–4, T–5
FR-76	3 propagation steps; $\Phi = 40$; dinitrobenzene intercepts the chain-carrying radical anion

FR-77	1-bromo-4-chlorobenzene; 4-bromobiphenyl, 0.41 g; meta
FR-78	12, 14, 54 and 7 lines (total 87); a(D) ≈ 0.35 mT
FR-79	(a) K ≈ 8.2×10^{-5} M; (b) ≈ 18 %
FR-80	1,2,2-trimethylpyrrolidine; the intermediate is 4-chloro-N,4-dimethylpentan-1-amine; six-membered TS
CE-1	6; 2 non-bonding
CE-2	(A), (C), (D)
CE-3	singlet: one (cis-1,1-dibromo-2,3-dimethylcyclopropane, meso); triplet: two (cis and trans)
CE-4	salicylaldehyde (major) and p-hydroxybenzaldehyde (minor); electrophile $:CCl_2$
CE-5	(a) and (c)
CE-6	$:CF_2 > :CCl_2 > :CBr_2 > :CI_2$
CE-7	(a) ICH_2ZnI (carbenoid) → norcarane; (b) $:CBr_2$ → 7,7-dibromonorcarane; (c) $:CH_2$ → norcarane (plus C–H insertion products)
CE-8	propanoic acid, CH_3CH_2COOH
CE-9	(D)
CE-10	$CHCl_3 \rightarrow CCl_3^- \rightarrow :CCl_2 \rightarrow HCCl(OH)\ldots \rightarrow HCOO^-$
CE-11	π-donation and electronegativity
CE-12	two π-donating N atoms and bulky N-substituents
CE-13	(a) cis-1,2-dimethylcyclopropane only (plus C–H insertion products); (b) mixture of cis and trans; (c) trans-1,1-dichloro-2,3-dimethylcyclopropane (stereospecific)
CE-14	2 products: n-butane : isobutane = 6 : 2.4 = 71 : 29 (statistical would be 75 : 25)
CE-15	reversible carbanion formation followed by rate-limiting loss of Cl^-
CE-16	$:CF_2 \approx :C(OCH_3)_2 > :CHF > :CH_2$ (triplet ground) > $:CPh_2$ (triplet ground, widest angle)
CE-17	singlet carbene that relaxes (intersystem crossing) to the triplet before reacting
CE-18	(a) salicylaldehyde (2-hydroxybenzaldehyde); (b) ion-pair / coordination of Na^+ with the phenoxide and carbene directs ortho; (c) $:CCl_2$; the intermediate is the dichloromethyl cyclohexadienone which loses Cl^- to give the o-quinone methide-type chloride that is hydrolysed
CE-19	(a) phenylacetic acid; (b) ethyl phenylacetate; (c) cyclopentanecarboxylic acid
CE-20	pK_a threshold for protonating CH_2N_2
CE-21	(b), (c), (d)
CE-22	5
CE-23	triplet carbenes; carbenoids of different type
CE-24	1,2-cycloheptadiene (strained cyclic allene, via the carbenoid and the cyclopropylidene), trapped as a dimer or cycloadduct
CE-25	$:CCl_2 + 2\ HO^- \rightarrow CO + 2\ Cl^- + H_2O$ (and $HCOO^-$ with more hydroxide)
CE-26	1 constitutional isomer (1,1-dichloro-2-vinylcyclopropane, as a racemate); the bis-adduct has 3 stereoisomers (R,R; S,S; meso)
CE-27	$CCl_3^- \rightarrow :CCl_2$; 7,7-dichloronorcarane
CE-28	see table
CE-29	Bamford–Stevens (NaOMe, Δ): more substituted alkene (carbene, 1,2-H shift, thermodynamic: 1-methylcyclohexene); Shapiro (2 RLi): less substituted alkene (3-methylcyclohexene) via the less hindered vinyl-lithium
CE-30	both true; the reason explains the assertion
CE-31	retention

CE-32	a bicyclic azulenone (5–7 fused) via carbene attack on the phenyl ring
CE-33	$CHCl_2^-$ is a less stable carbanion than CCl_3^-
CE-34	(a) butylcyclopropane (selective, carbenoid); (b) butylcyclopropane plus C–H insertion products (isomeric heptenes and alkanes)
CE-35	triplet
CE-36	1,1-dibromo-trans-2,3-diphenylcyclopropane; Ag^+ ionises it to a cyclopropyl cation that ring-opens (disrotatory) to a substituted allyl cation (→ 2-bromo-1,3-diphenylallyl derivatives)
CE-37	tetramethylethylene > 2-methyl-2-butene > cis-2-butene > 1-hexene > ethylene
CE-38	(a) one product, meso cis-1,2-dimethylcyclopropane; (b) trans-1,2-dimethylcyclopropane as a racemate (two enantiomers)
CE-39	diazoacetone, then propanoic acid
CE-40	exo-3,3-dichlorotricyclo[3.2.1.$0^{2,4}$]octane
CE-41	thiazol-2-ylidene (NHC); the Breslow intermediate; the polarity of the carbonyl carbon (umpolung)
CE-42	2-hydroxy-5-methylbenzaldehyde; 4-(dichloromethyl)-4-methylcyclohexa-2,5-dienone
CE-43	3-chloropyridine; a 6,6-dichloro-2-azabicyclo[3.1.0]hexene (dichlorocyclopropane)
CE-44	3-methylbuta-1,2-diene
CE-45	bicyclo[3.1.0]hexan-2-one
CE-46	they give a carbene that inserts into C–H/X–H bonds indiscriminately
CE-47	6 (a, b, c, d, e, h)
CE-48	x = 1, y = 3
CE-49	4 stereoisomers (two diastereomers, each as a pair of enantiomers)
CE-50	Pair 1: (a); Pair 2: (d); Pair 3: (c); Pair 4: (b)
CE-51	2-hydroxy-5-methylbenzaldehyde (normal); 4-(dichloromethyl)-4-methylcyclohexa-2,5-dienone (abnormal)
CE-52	3 (CH_3NH_2, $C_6H_5NH_2$, $(CH_3)_3CNH_2$)
CE-53	(i) cyclopropanation of the C2=C3 (allylic) alkene; (ii) epoxidation of the C6=C7 alkene
CE-54	(*S*)-3-methylpentanoic acid; intermediate: ketene (from Wolff rearrangement)
CE-55	(a) ethyl cyclohepta-2,4,6-triene-1-carboxylate (Buchner ring expansion); (b) a 5–7 fused hydroazulenone; (c) π-acceptors (e.g. 7,7-dicyano)
CE-56	tetramethylallene (3 ^{13}C signals); 1,2-cyclononadiene (chiral allene)
CE-57	A: 3-methylcyclohexene; B: 6-methylcyclohex-1-ene-1-carbaldehyde; C: 1-methylcyclohexene (major)
CE-58	A, B, C, D
CE-59	(i) poly(1,3-cyclopentylenevinylene); (ii) *cis*-1,3-divinylcyclopentane (meso)
CE-60	(a) 1-phenylpentane-1,4-dione; (c) benzoin
CE-61	ethyl 3-methyl-2-(phenylsulfanyl)pent-4-enoate; 4 stereoisomers (2 racemic diastereomers)
CE-62	(a) 3,3-dimethylcyclopentanone; (b) (*R*)-3-ethyl-3-methylcyclopentanone
CE-63	(a) 1-methoxy-4-(phenylethynyl)benzene; label on the carbon bonded to Ph means the anisyl group migrated; (b) 4-phenylbut-1-yne via alkylidene carbene and 1,2-H shift
CE-64	(a) 3-chloropyridine; (b) C-3; (c) 3-chloroquinoline; (d) pyrrole-2-carbaldehyde
CE-65	(a) bicyclo[3.1.0]hexan-2-one, 2 stereoisomers (enantiomers); (c) 3-methylcyclopentanone
CE-66	B = methyl indane-2-carboxylate; ketene = indan-2-ylideneketene; excluded isomer has 11 signals

CE-67	(a) D on the carbinol carbon only: PhCO–CD(OH)Ph; (b) PhCO–CH(OH)Ar is unlabelled; ArCO–CD(OH)Ph carries the D
CE-68	(a) 2,3-diethyl-4-methoxynaphthalen-1-ol; (b) 2-butyl-4-methoxynaphthalen-1-ol
NI-1	6; both (lone pair + empty/half-filled p orbitals); mainly electrophilic
NI-2	(a) acyl nitrene (nitrenoid; concerted isocyanate formation), (b) same, (c) alkylnitrene, (d) dichlorocarbene
NI-3	ethanamine (ethylamine), one carbon less
NI-4	aniline; benzylamine
NI-5	retention of configuration (the amine is (S)-sec-butylamine in the CIP sense)
NI-6	(D)
NI-7	N_2; Br^-; carboxylate; N_2
NI-8	3H-azepine (2-diethylamino-3H-azepine); ring-expanded didehydroazepine
NI-9	singlet: stereospecific; triplet: loss of stereochemistry
NI-10	two degenerate p orbitals each hold one electron (Hund's rule)
NI-11	(a) CH_3NH_2; (b) t-$BuNH_2$; (c) benzylamine; (d) cyclohexylamine; (e) 1,4-diaminobutane (putrescine)
NI-12	anthranilic acid (2-aminobenzoic acid)
NI-13	(a) (i) $SOCl_2$; NaN_3; Δ; H_2O (Curtius), or (ii) $SOCl_2$; NH_3; Br_2/KOH (Hofmann); (b) reduce acid or its amide to the alcohol/amine: $CH_3CH_2COOH \rightarrow CH_3CH_2CONH_2 \rightarrow LiAlH_4 \rightarrow CH_3CH_2CH_2NH_2$
NI-14	(R)-1-phenylpropylamine (retention); no racemisation
NI-15	(a) only (and arguably (f) gives NH_3/HNCO but not a meaningful amine)
NI-16	(a) ε-caprolactam; (b) acetanilide (phenyl migrates); (c) aniline
NI-17	same product (ε-caprolactam); Beckmann
NI-18	E: N-(4-methylphenyl)acetamide (aryl migrates); Z: N-methyl-4-methylbenzamide
NI-19	(a) no (concerted Curtius); (b) partly yes; (c) yes (singlet then triplet); (d) no (copper–imido, nitrenoid); (e) yes (arylnitrene, or nitrenoid with phosphorus)
NI-20	triplet phenylnitrene
NI-21	A–Q; B–P; C–R; D–S
NI-22	A = 2-methylbutanamide; B = butan-2-amine
NI-23	the amide N–H must be removed to give the anion; the neutral N-bromoamide is stable
NI-24	carbazole
NI-25	cis → cis-aziridine; trans → trans-aziridine
NI-26	triplet; azepine arises from singlet-derived ring expansion
NI-27	see reasoning
NI-28	methyl N-benzylcarbamate
NI-29	4
NI-30	conjugation with the arene lowers the TS energy; the aryl π system offers cyclisation, the alkyl group offers a migrating H
NI-31	(S)-1-phenylethylamine
NI-32	$RCONH_2 + Br_2 + 4\ NaOH \rightarrow RNH_2 + Na_2CO_3 + 2\ NaBr + 2\ H_2O$; 4 mol
NI-33	carbazole via an aryl nitrene (insertion into the adjacent ring C–H)
NI-34	N-ethoxycarbonyl-7-azabicyclo[4.1.0]heptane (aziridine) and allylic C–H insertion product
NI-35	R–NH–Boc via acyl azide and isocyanate

NI-36	p-aminophenol via a nitrenium ion (Bamberger rearrangement)
NI-37	p-phenylenediamine
NI-38	2-phenyl-1-tosylaziridine; opening at the benzylic carbon
NI-39	6 ; (S)-butan-2-amine
NI-40	10 (8 NaOH + 2 Br2); butane-1,4-diamine
NI-41	A, B, C, E
NI-42	E: acetanilide → aniline + acetic acid; Z: N-methylbenzamide → methylamine + benzoic acid
NI-43	pentan-2-amine; major pent-1-ene (C5)
NI-44	anthranilic acid (C7), β-alanine (C3), 4-aminobutanoic acid (C4); sum = 14
NI-45	2-phenylindole; 2 equiv P(OEt)3, giving 2 OP(OEt)3 (triethyl phosphate)
NI-46	2-(diethylamino)-3H-azepine, C10H16N2; 5 sp3 carbons
NI-47	No; spatial arrangement retained but descriptor changes S → R
NI-48	P–2, Q–1, R–3, S–4, T–5 (6 matches none)
NI-49	(S)-7-methylazepan-2-one; 3-methylazepan-2-one; nylon-6 unit –[NH(CH2)5CO]–
NI-50	imine Ph(4-O2NC6H4)C=N-Ph (N-phenyl imine); then 4-nitrobenzophenone + aniline
NI-51	(i) 4-aminophenol; (ii) 4-methoxyaniline; phenylnitrenium ion, charge on 4 atoms
NI-52	Pair 1: A; Pair 2: A; Pair 3: D
NI-53	A cis-fused N-tosylaziridine; B trans-azido sulfonamide; C trans-N-tosyl-1,2-diaminocyclohexane (racemic, 2 stereoisomers); no free nitrene
NI-54	(i) phenylacetaldehyde (C8); (ii) methyl (E)-styrylcarbamate, PhCH=CH–NHCO2CH3 (C10)
NI-55	oxime; caprolactam; azepane; N,N-dimethylhex-5-en-1-amine; hexa-1,5-diene; 3 CH3I; NMe3
NI-56	P 4,4-dimethyl-1,2,3-oxathiazinane 2,2-dioxide (6-ring); Q 4-azido-2-methylbutan-2-amine; R 3-methylbutane-1,3-diamine
NI-57	X = (S)-3-methylpentanoic acid (C6); amine = (S)-2-methylbutan-1-amine (C5); direct Curtius gives (S)-butan-2-amine (C4)
NI-58	4 constitutional isomers; 6 stereoisomers; major: ethyl N-(2-methylbutan-2-yl)carbamate
NI-59	K propiophenone; A (E)-oxime; B N-phenylpropanamide; C anilinium chloride; D propanoic acid; 7 13C signals
NI-60	(1S,2R)-trans-2-phenylcyclopropylamine; C-2 descriptor flips S → R though geometry is retained
BZ-1	(B), (C), (D)
BZ-2	benzyne → aniline; no reaction for 2,6-dimethylbromobenzene
BZ-3	(a) 2; (b) 3; (c) 2
BZ-4	m-anisidine (3-methoxyaniline)
BZ-5	(i) and (ii)
BZ-6	Br > I > Cl ⩾ F (Roberts)
BZ-7	1,4-dihydro-1,4-epoxynaphthalene
BZ-8	sp^2; ≈ 120–127° (not 180°)
BZ-9	addition–elimination (S_NAr), not benzyne
BZ-10	6π (aromatic) + 2 electrons in the extra in-plane bond; neutral
BZ-11	(b) and (d)
BZ-12	(a) 2 (o- and m-); (b) 3 (o-, m-, p-); (c) 2 (m- and p-)

BZ-13	o → m-anisidine (only); m → m-anisidine (only); p → m- and p-anisidine (≈ 1 : 1)
BZ-14	triptycene (+ CO_2 + N_2 + 2 isoamyl alcohol)
BZ-15	both correct; the reason explains the assertion
BZ-16	o-chloronitrobenzene: S_NAr (addition–elimination); chlorobenzene: no reaction (needs benzyne conditions, $NaNH_2$, or 623 K/300 atm)
BZ-17	(a) 1,4-epoxy-1,4-dihydronaphthalene; (b) benzonorbornadiene; (c) 1,2,3,4-tetraphenylnaphthalene (loses CO); (d) naphthalene (loses CO_2)
BZ-18	m-fluoroaniline
BZ-19	3-bromopyridine forms 3,4-pyridyne; 2-halopyridine reacts by S_NAr
BZ-20	no ortho H: no benzyne; but strongly activated ring: S_NAr
BZ-21	C–F bond is strong and F^- is a poor leaving group, although the ortho-H is more acidic
BZ-22	o-chloro: m-CF_3-aniline only; m-chloro: m-CF_3-aniline predominantly
BZ-23	phenylacetone (1-phenylpropan-2-one)
BZ-24	o-fluorophenylmagnesium bromide, then benzyne (loss of MgBrF); 1,4-epoxy-1,4-dihydronaphthalene
BZ-25	3 possible (o-, m-, p-); m-anisidine is obtained
BZ-26	p-benzyne is a σ,σ-diradical; ortho-benzyne is a strained electrophilic alkyne
BZ-27	(a) aniline; (b) anisole; (c) biphenyl; (d) N-methyl-1,4-dihydronaphthalen-1,4-imine (Diels–Alder adduct)
BZ-28	NO_2 stabilises the Meisenheimer anion; CH_3 cannot, so the strong base removes the ortho H
BZ-29	3,4-pyridyne from 3-bromopyridine gives both; 4-bromopyridine reacts partly by S_NAr
BZ-30	1 benzyne, 2 amines
BZ-31	1-naphthol
BZ-32	(a) o-bromophenyllithium; (b) benzyne (LiBr lost)
BZ-33	3 signal types; benzyne + anthracene
BZ-34	S_NAr versus elimination–addition
BZ-35	1,4-didehydrobenzene (p-benzyne) diradical; benzene
BZ-36	triphenylene, $C_{18}H_{12}$
BZ-37	7
BZ-38	about 1 : 1 (C-1 : C-2); a direct substitution would give 100 : 0
BZ-39	(a)
BZ-40	P–1, Q–2, R–3, S–4, T–5
BZ-41	A = 1,4-dihydro-1,4-epoxynaphthalene (achiral, 1 stereoisomer); B = 1-naphthol
BZ-42	P–i, Q–ii, R–iii, S–iv, T–i
BZ-43	both give 3-methoxyaniline; 2-bromoanisole: cine, 3-bromoanisole: ipso
BZ-44	T = triptycene, $C_{20}H_{14}$; 3 ^1H signals (2H : 6H : 6H) and 4 ^{13}C signals
BZ-45	(i) 1,2,3,4-tetraphenylnaphthalene, $C_{34}H_{24}$ + CO; (ii) naphthalene, $C_{10}H_8$ + CO_2
BZ-46	ene reaction; P = (2-methylallyl)benzene; minor: 1,1-dimethylbenzocyclobutene ([2+2]); 1 D on the ring
BZ-47	(i) 1-benzylbenzotriazole (3 ring N); (ii) 2-methyl-3-phenyl-2,3-dihydro-1,2-benzisoxazole (ring N and O, 2 heteroatoms)
BZ-48	A, B and C
BZ-49	3-bromopyridine (3,4-pyridyne); n = 3

BZ-50	6-(dialkylamino)-1-methylindole (attack at C-6)
BZ-51	A, B, C and D
BZ-52	p-benzyne (5,6,7,8-tetrahydronaphthalene-1,4-diyl); tetralin $C_{10}H_{12}$ and benzene
BZ-53	4: 2,3-, 3,4-, 2,4- and 3,5-dimethylaniline
BZ-54	(a) $C_{12}H_8$ and $C_{18}H_{12}$; (b) two: 2,6- and 2,7-dimethylbiphenylene
BZ-55	X = ethyl 2-(2-acetylphenyl)acetate (C–C σ-bond insertion)
BZ-56	D = 1,3-diphenylisobenzofuran; E = its benzyne [4+2] adduct (9,10-diphenyl-9,10-epoxy-9,10-dihydroanthracene); F = 9,10-diphenylanthracene
BZ-57	4, 2, 5, 3, 1; key intermediate p-benzyne; Myers–Saito gives an α,3-didehydrotoluene (σ,π) biradical
BZ-58	aryne: C–X cleavage is rate-limiting-type; SNAr: addition is rate-limiting; fluorobenzene: fast reversible deprotonation, slow loss of F^-
RI-1	1-methoxycyclohexa-1,4-diene; cyclohexa-2,5-diene-1-carboxylic acid
RI-2	ketyl radical anion; pinacol; pinacolone
RI-3	(a) and (d)
RI-4	the molecular radical cation; fragmentation gives the more stable carbocation + radical
RI-5	(a) 1-methylcyclohexa-1,4-diene; (b) 1,4-dimethylcyclohexa-1,4-diene; (c) cyclohexa-2,5-diene-1-carboxylic acid; (d) 1-methoxycyclohexa-1,4-diene
RI-6	(a) (E)-hex-3-ene; (b) (Z)-hex-3-ene; (c) the terminal alkyne is deprotonated, giving sodium hex-1-ynide (not reduced)
RI-7	4 (naphthalene, benzophenone, anthracene, acetone)
RI-8	(a) deep blue/purple; indicator of dryness and O_2-free conditions in solvent stills; (b) the colour is discharged and benzhydrol/benzopinacol form
RI-9	benzene > hexane > ethanol > 2-methylpentan-2-ol
RI-10	m/z 43: CH_3CO^+ by α-cleavage; m/z 58: McLafferty rearrangement (loss of ethene)
RI-11	one chlorine (^{35}Cl : ^{37}Cl ≈ 3 : 1); loss of Cl• gives $C_3H_7^+$ (m/z 43)
RI-12	the rate-limiting step is C–X cleavage of the radical anion, not nucleophilic addition; no aryne is involved
RI-13	$S_{RN}1$ radical chain
RI-14	(a) 2,3-diphenylbutane-2,3-diol (pinacol); (b) (E)-stilbene
IP-1	(a) carbocation (after a hydride shift); (b) benzylic radical; (c) benzyne; (d) carbanion CH_3^-; (e) isocyanate via acyl nitrenoid; (f) dichlorocarbene (→ phenyl isocyanide)
IP-2	carbocation (6), carbene (6), nitrene (6) and free radical (7)
IP-3	triplet carbene = triplet nitrene (2) > free radical (1) > carbocation = singlet carbene (0)
IP-4	3 (cation, carbene, nitrene)
IP-5	CCl_3^- (carbanion) → :CCl_2 (carbene) → o-dienone carbanion → o-(dichloromethyl)phenoxide → $ArCH(OH)_2$ → ArCHO
IP-6	benzyl bromide
IP-7	A–S,Q; B–R; C–Q; D–P,S
IP-8	cation: hydride abstraction (Ph_3C^+); radical: NBS or Cl_2/hν; anion: n-BuLi/KNH_2
IP-9	(a) racemic substitution product (planar cation); (b) racemic bromide (planar radical); (c) retention: (R)-1-phenylethylamine
IP-10	$CHCl_3$ + KOH (α-elimination)
IP-11	(i) 2-bromo-2-methylbutane via the t-amyl cation (methyl shift); (ii) neopentane (no rearrangement)

IP-12	S_NAr (addition–elimination) vs benzyne (elimination–addition)
OR-1	$CH_3COOH < ClCH_2COOH < FCH_2COOH < Cl_2CHCOOH < Cl_3CCOOH$
OR-2	p-cresol < phenol < m-nitrophenol < p-nitrophenol < 2,4-dinitrophenol
OR-3	p-OMe < p-CH_3 < benzoic < m-OMe < p-NO_2
OR-4	aniline < NH_3 < $(CH_3)_3N$ < CH_3NH_2 < $(CH_3)_2NH$
OR-5	pyrrole < aniline < pyridine < imidazole < piperidine
OR-6	solvation of the smaller anion versus polarisability of the larger alkyl groups
OR-7	amide < ester < anhydride < acid chloride
OR-8	acetophenone < acetone < benzaldehyde < acetaldehyde < formaldehyde
OR-9	nitrobenzene < chlorobenzene < benzene < toluene < anisole
OR-10	2,3-dimethylbut-2-ene > 2-methylbut-2-ene > trans-2-butene > cis-2-butene > 1-butene
OR-11	$–NO_2$ > –CN > –F > –Br > $–OCH_3$ > $–C_6H_5$
OR-12	HO^- < F^- < Cl^- < Br^- < I^-; protonation turns HO^- into neutral H_2O
OR-13	S_N2: $(CH_3)_3CBr$ < $(CH_3)_2CHBr$ < CH_3CH_2Br < CH_3Br; S_N1: the reverse (CH_3Br < 1° < 2° < 3°)
OR-14	–M at para delocalises the lone pair more; aryl N lone pair is delocalised into the ring
OR-15	4
LG1	D > C > B > A
LG2	A > B > C > D
LG3	A > B > C > D
LG4	C > D > B > A
LG5	A > B > C
LG6	D > C > B > A
LG7	A > B > C > D > E
LG8	A > C > B
LG9	A > B > C > D
LG10	A > B > C
LG11	A > B > C > D
LG12	A > B > C
LG13	A > B > C > D
LG14	A > B > C > D
LG15	A > B > C > D
R1	5 (a, d, e, h, j)
R2	4 (a, b, d, e)
R3	(i) 2 (ii) 1
R4	2
R5	3
R6	4 (a, d, e, h)
R7	4 (c, d, f, g)
R8	4 (a, b, e, g)